Do Now · retrieval

Tree traversal: post-order and breadth-first

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a constructor do?

Q2

What is in-order traversal order?

Q3

What does FIFO stand for?

Q4

A root R has only left child L. What is post-order?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Depth-first post-order — Visit an entire left subtree, then right subtree, and finally the current node.

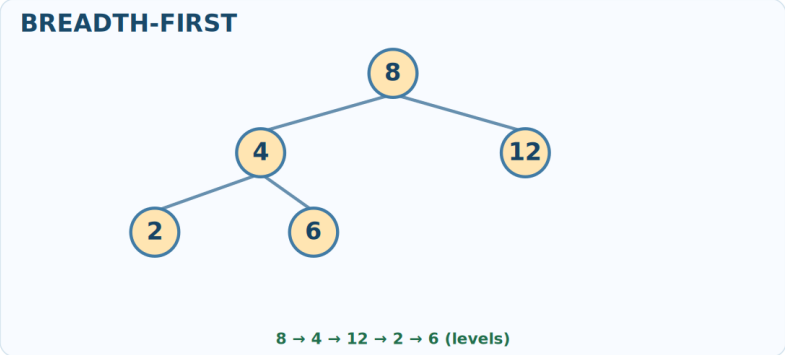
Core idea 2

Breadth-first level order — Start with the root in a queue. Dequeue one node, visit it and enqueue each of its children.

Core idea 3

Compare output — Same structure yields different orders: 2,6,4,12,8 versus 8,4,12,2,6.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

post-order — Traverse left, right, then root.

breadth-first — Visit vertices level by level.

FIFO queue — First-in-first-out structure used by BFS.

leaf — Node without children.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

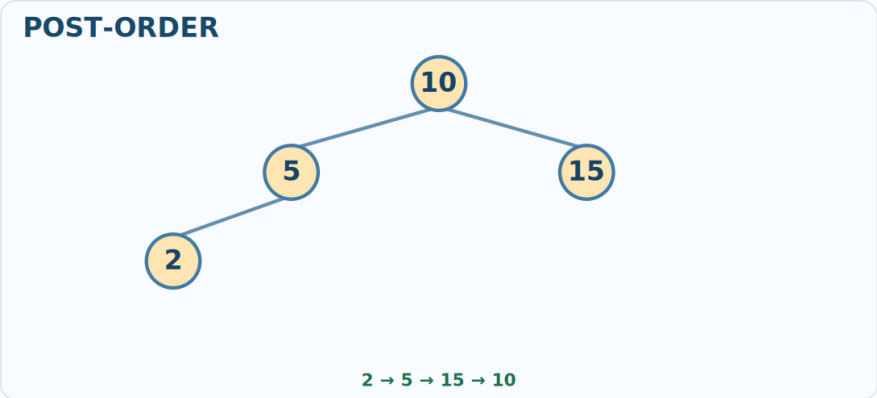
Find post-order traversal of root 10, children 5 and 15, with 5 having left child 2.

Follow the worked method

1. Predict first — PostOrder(10) calls left(5), then left(2). At node2 both null-child calls return, so print2; return to node5 and print5.

2. Model the reasoning — Return to10, traverse right child15 and print15, then print10. Output 2,5,15,10: the print follows both child calls. Node has Key, Left and Right fields.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Visit root 20, enqueue 10 and 30.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A preorder routine prints node.Key, then traverses Left and Right (with a null guard). Move one statement to make it post-order. Give the result for root6 with children3 and9.

Question 2

Tree root 12, left 5 with children 2 and 8, right 17. Give breadth-first and post-order traversals.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write pseudocode to traverse a binary tree breadth-first. Explain why a queue rather than a stack is used.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Breadth-first traversal normally uses a stack to ensure earliest-discovered nodes are visited first.

Use key vocabulary from memory

1. Explain post-order in your own words.

2. Explain breadth-first in your own words.

3. Explain FIFO queue in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.